

# DNA sequencing, FASTQ format, tools.

BFX Workshop Week 3

Chris Miller

**Many slides adapted from:**

**Applied Computational Genomics** <https://github.com/quinlan-lab/applied-computational-genomics>

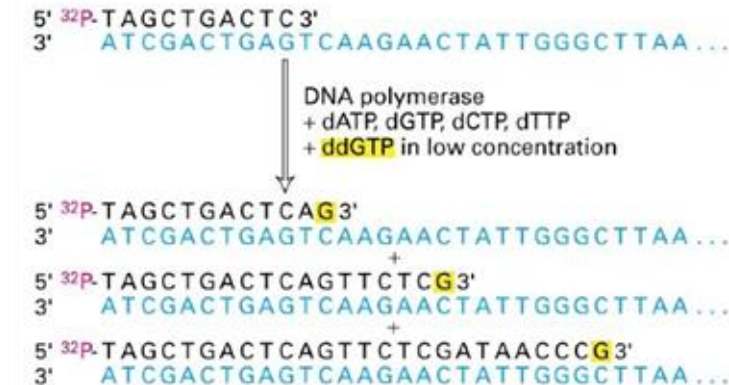
**Aaron Quinlan**



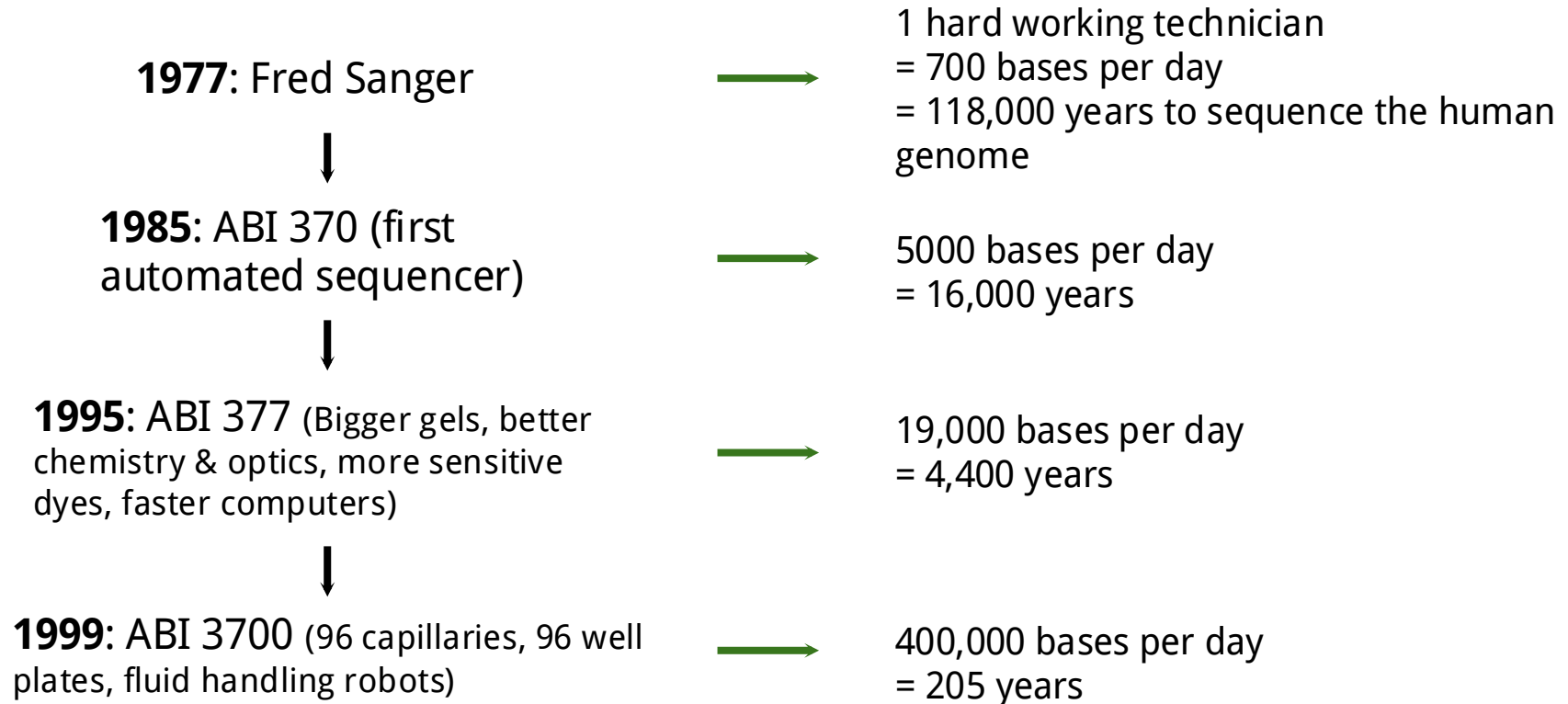
# How to sequence a human genome: Sanger method

## Key points:

- 1) sequencing by synthesis (not degradation)
- 2) primers hybridize to DNA
- 3) polymerase + dNTPS + ddNTP terminators at low concentration
- 4) 1 lane per base, visually interpret ladder



# Sanger sequencing: technological advances



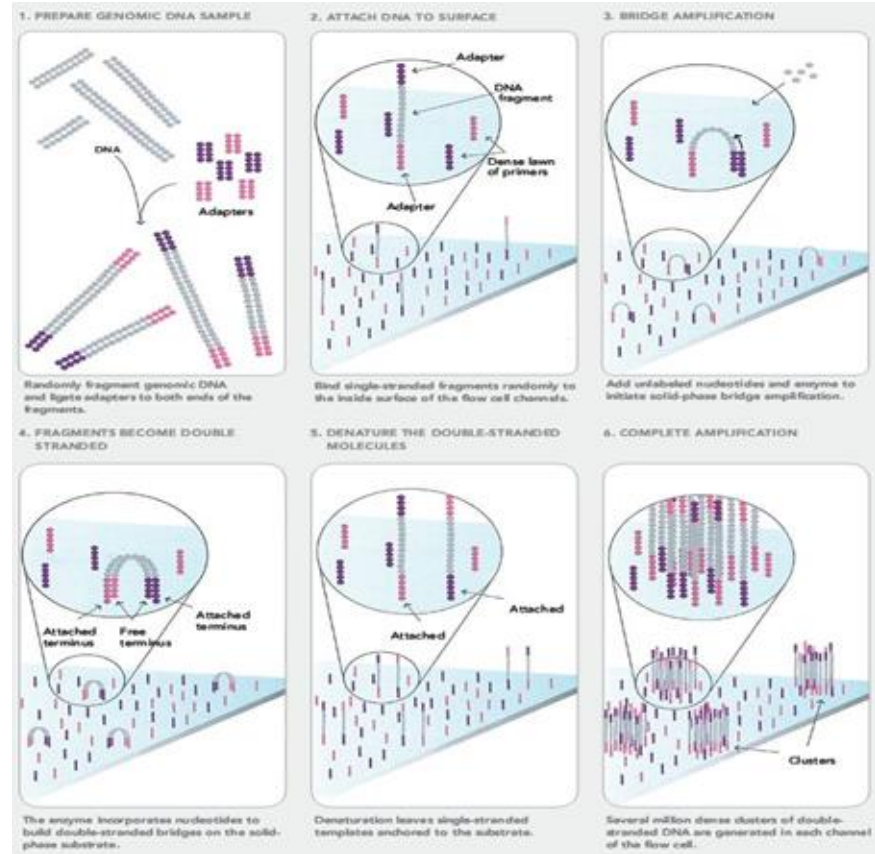
# The next wave of DNA sequencing technologies

## whiz-bang terms

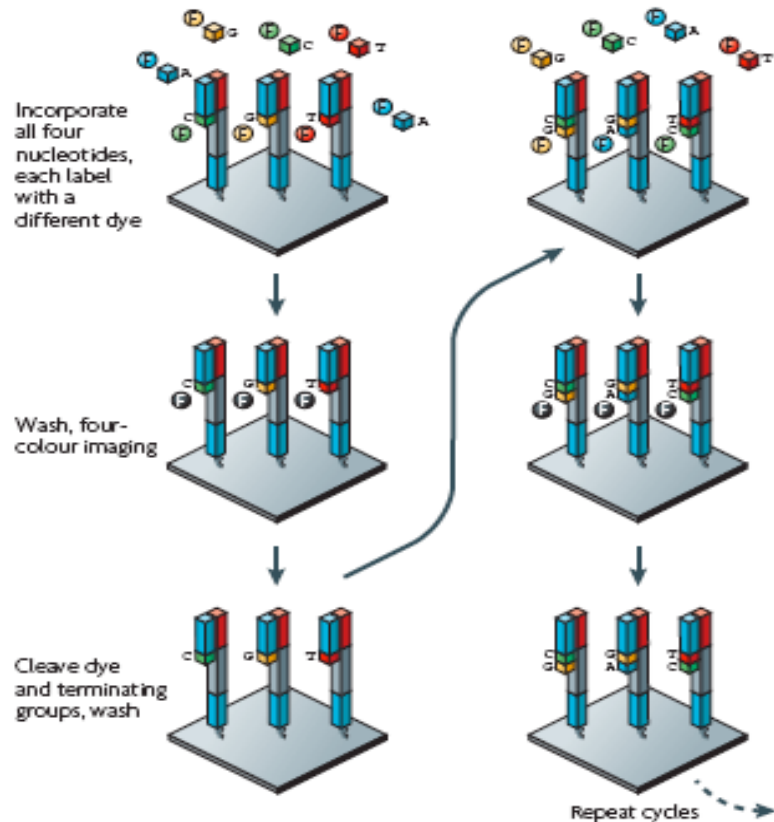
- “Massively parallel” sequencing
  - “High-throughput” sequencing
  - “Ultra high-throughput” sequencing
  - “Next generation” sequencing (NGS)
  - “Second generation” sequencing
- **2005: 454 (Roche)**
  - **2006: Solexa (Illumina)**
  - **2007: ABI/SOLiD (Life Technologies)**
  - **2010: Complete Genomics**
  - **2011: Pacific Biosciences**
  - **2010: Ion Torrent (Life Technologies)**
  - **2015: Oxford Nanopore Technologies**

# Solexa (Illumina) sequencing (2006)

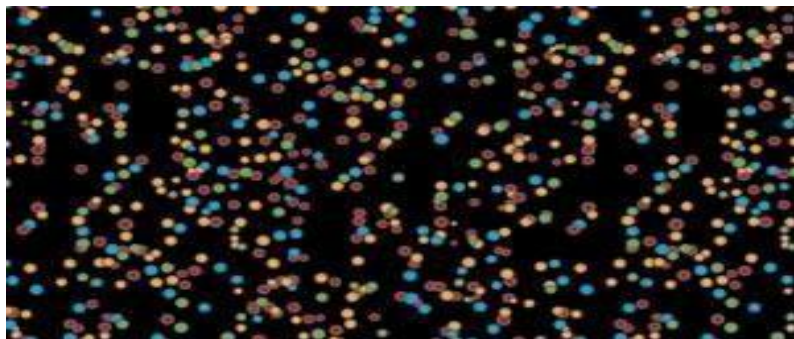
- PCR amplify sample (opt.)
- Immobilize and amplify single molecules on a solid surface
- Reversible terminator sequencing with 4 color dye-labelled nucleotides



# Cluster amplification by "bridge" PCR



4 different images merged



6 cycles w/ base-calling



# Illumina sequencing summary

## Advantages:

- . Best throughput, accuracy and read length for any 2nd gen. sequencer
- . Fast & robust library preparation

## Disadvantages:

- . Inherent limits to read length (practically, 150bp)
- . Some runs are error prone



Illumina X-plus

8-16 Tb per run  
64 human genomes per run  
48 hours

~\$400 per genome

# Paired-end sequencing:

## A molecular hack to sequence longer fragments

genomic DNA



Shear to desired length (~400bp)

DNA fragments

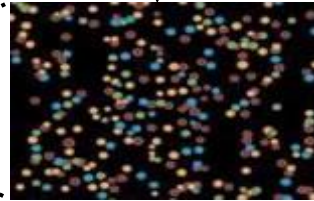


ligate adapters, size select

sequencing library



Illumina GA2



clusters on a flow-cell



millions to billions of paired-end reads (readpairs)

~150bp

~200bp

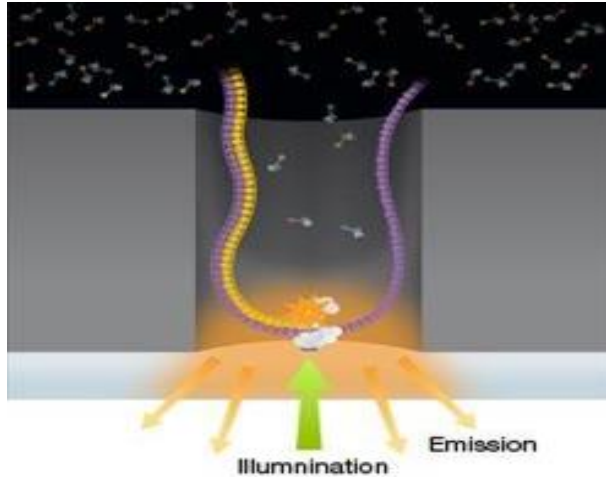
~150bp

5' GGTGTACGAATAGTTTCCTTTTACACTCCTTGACCATCCTAGC -----//----- GGACTGAAACTTCATCTGTCTTTATAGATATGCGTGCAGCAGC 5'

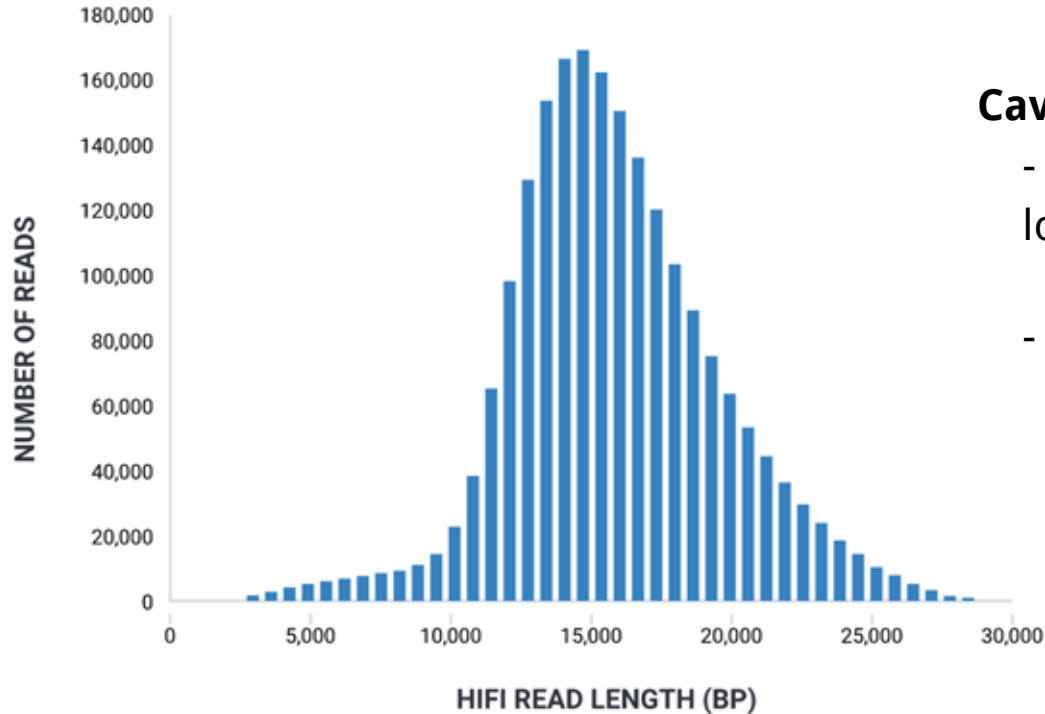
# Pacific Biosciences

## Key Points:

- 1 DNA molecule and 1 polymerase in each well (zero-mode waveguide)
- 4 colors flash in real time as polymerase acts
- Methylated cytosine has distinct pattern
- No *theoretical* limit to DNA fragment length



# Pacific Biosciences: long reads. Great for genome assembly

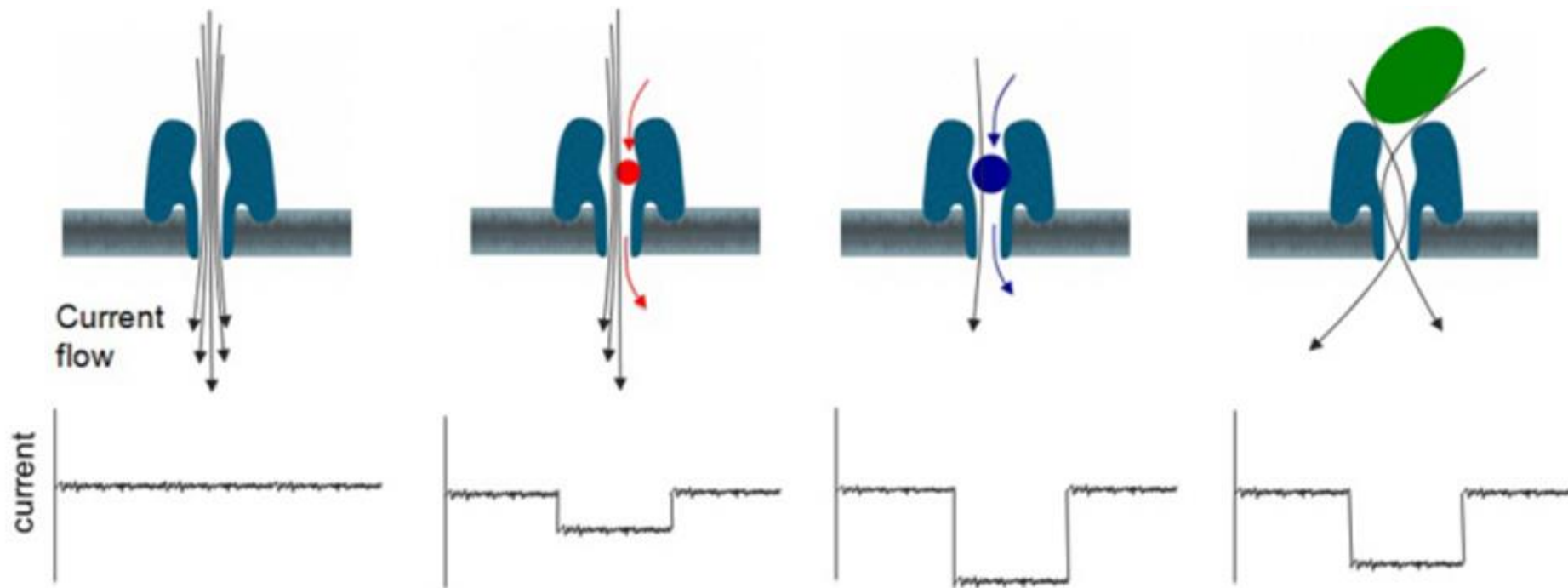


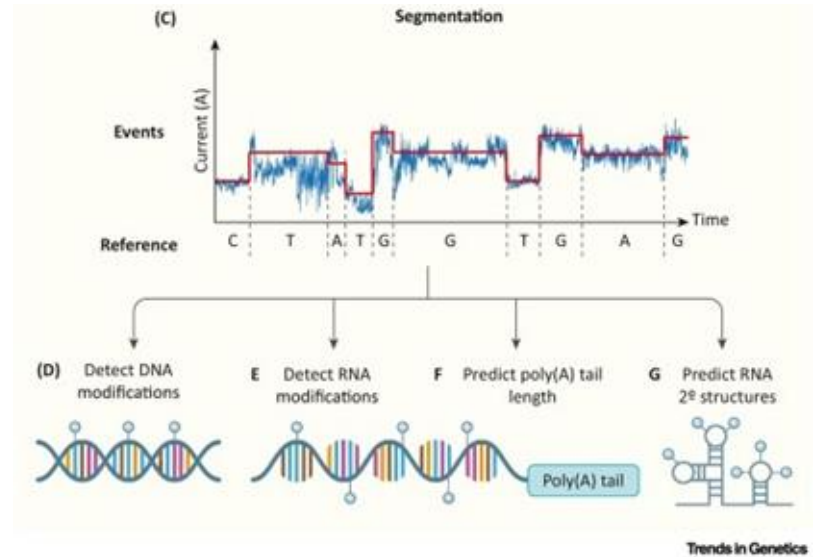
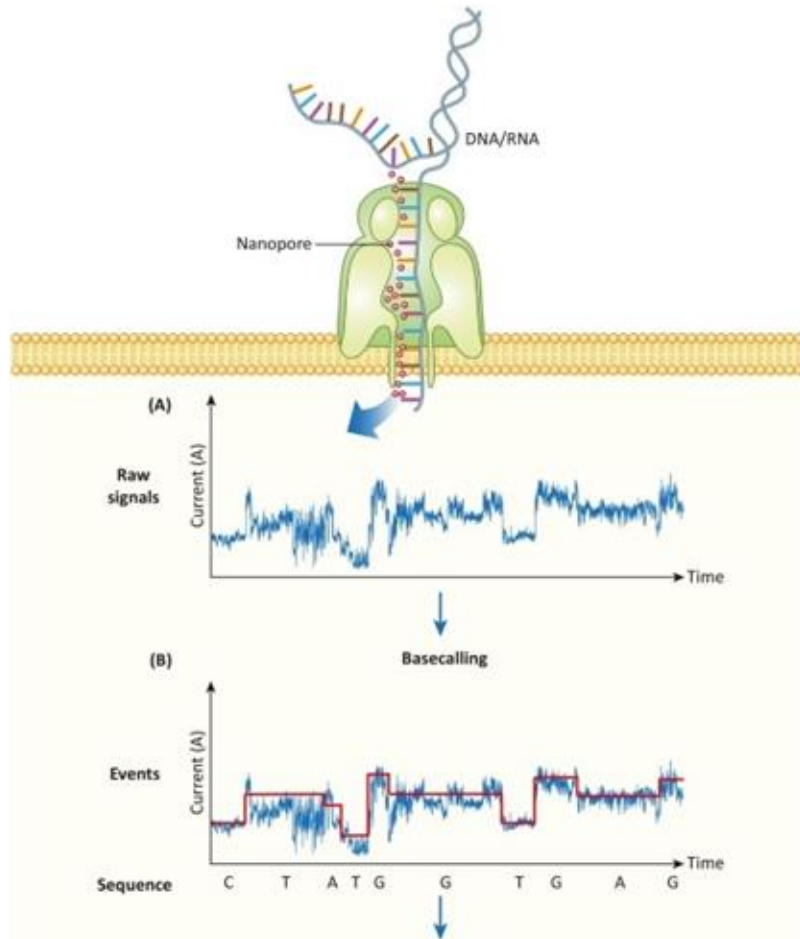
## Caveats:

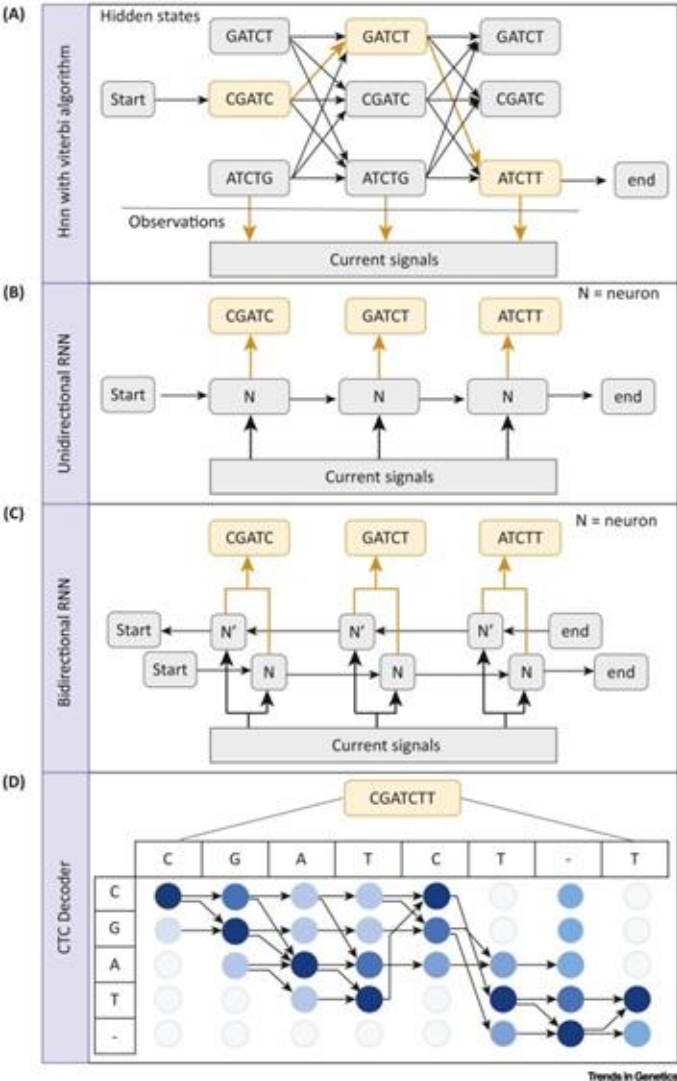
- higher error rate (1-2%), lower with Duplex runs
- lower throughput

About \$4,000 for a 30x human genome on the RevIO machine

# Oxford Nanopore Technologies







# Neural networks to translate signal into base calls

- Guppy (many versions)
- Dorado (v0.4, eventual guppy replacement)
- many others

Practically, that means that we can't yet throw away our raw signal intensities. (1 Tb or more per run)

# ONT sequence length distribution

## Legend

Basecalled Estimated

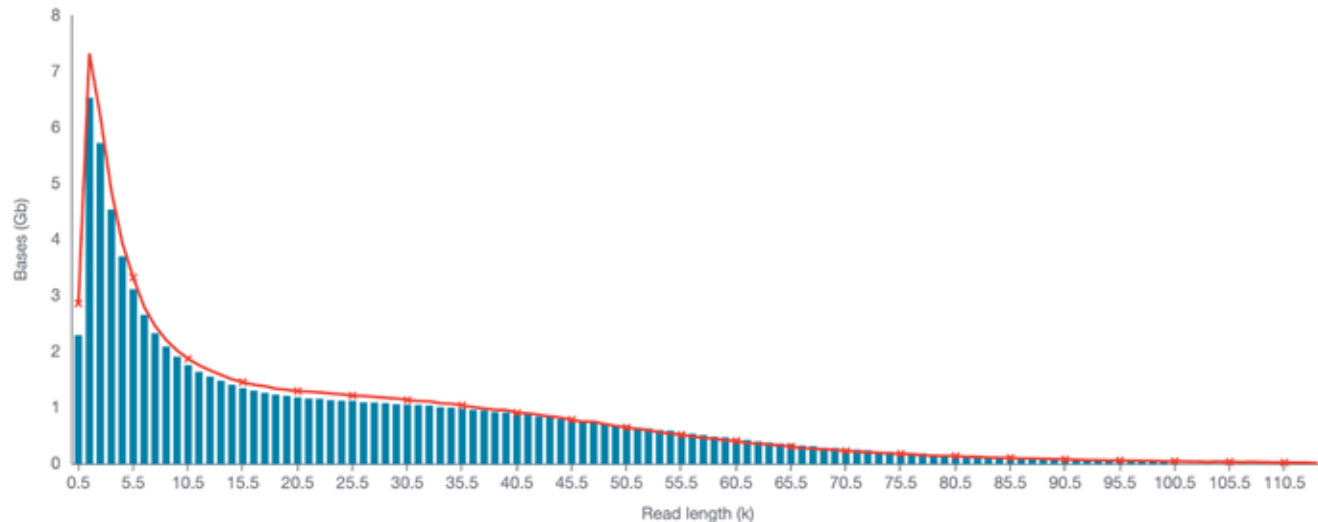
Estimated N50

17.75 kb

% Basecalled

100%

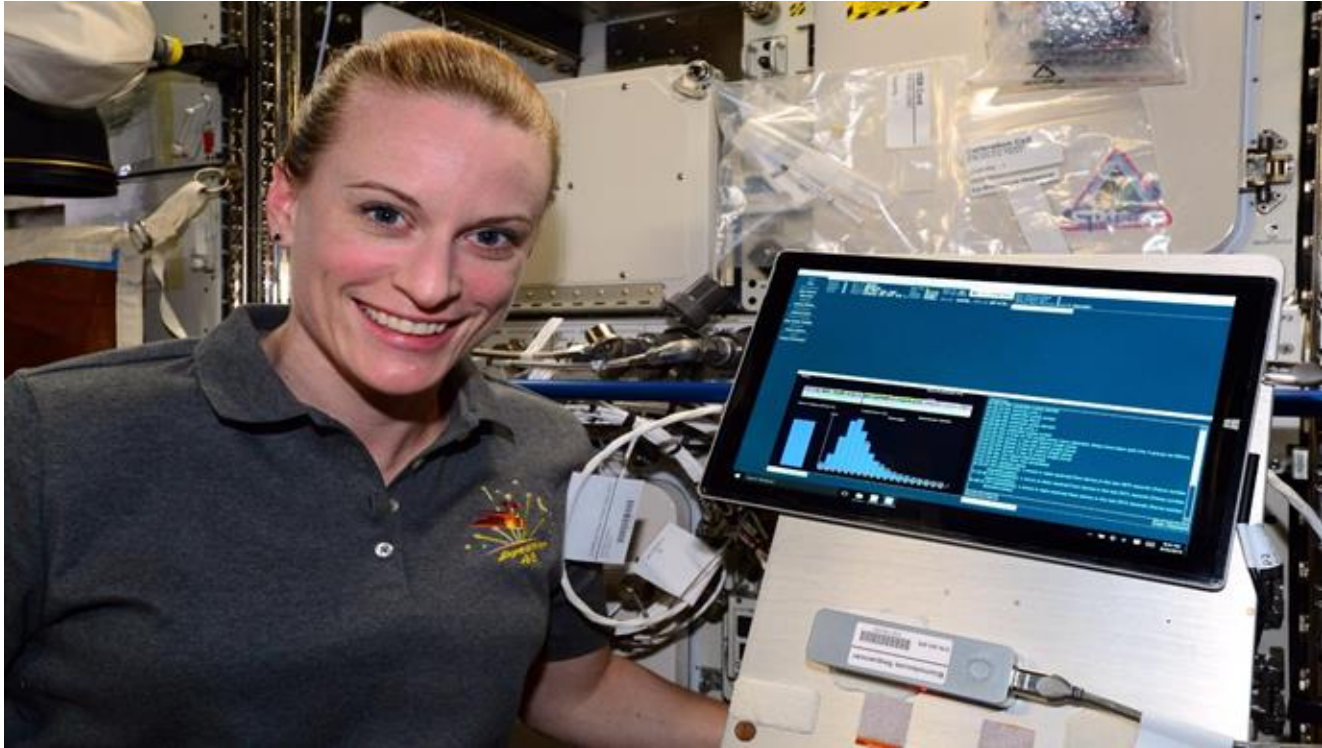
their relative amounts.



| Read length (kb) | Aggregated reads (Mb) |
|------------------|-----------------------|
| 100 - 164        | 886.98                |
| 164 - 228        | 36.06                 |
| 228 - 292        | 4.02                  |
| 292 - 344        | 0.35                  |

Recent run of a tumor sample

Nanopore sequencing is *extremely* portable



Kate Rubins sequencing DNA on the ISS

# Long reads. Great for genome assembly

## Single haplotype assembly of the human genome from a hydatidiform mole

Karyn Meltz Steinberg,<sup>1</sup> Valerie A. Schneider,<sup>2</sup> Tina A. Graves-Lindsay,<sup>1</sup> Robert S. Fulton,<sup>1</sup> Richa Agarwala,<sup>2</sup> John Huddleston,<sup>3,4</sup> Sergey A. Shiryev,<sup>2</sup> Aleksandr Morgulis,<sup>2</sup> Urvashi Surti,<sup>5</sup> Wesley C. Warren,<sup>1</sup> Deanna M. Church,<sup>6</sup> Evan E. Eichler,<sup>3,4</sup> and Richard K. Wilson<sup>1</sup>

<sup>1</sup>The Genome Institute at Washington University, St. Louis, Missouri 63108, USA; <sup>2</sup>National Center for Biotechnology Information, National Library of Medicine, National Institutes of Health, Bethesda, Maryland 20894, USA; <sup>3</sup>University of Washington, Seattle, Washington 98195, USA; <sup>4</sup>Department of Pathology, University of Washington, Seattle, Washington 98195, USA; <sup>5</sup>Department of Pathology, University of Washington, Seattle, Washington 98195, USA; <sup>6</sup>Personalis, Inc., Menlo Park, California 94025, USA

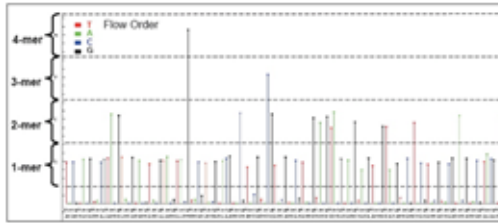
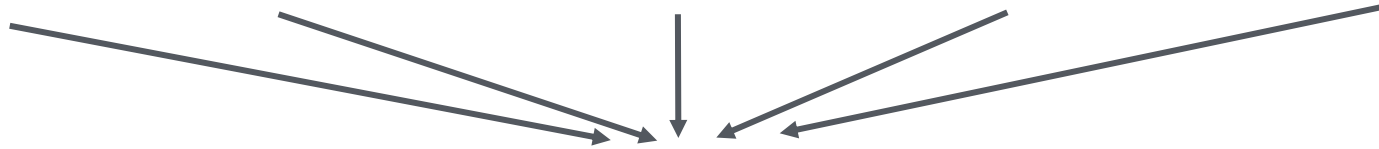
## The complete sequence of a human genome

SERGEY NURK, SERGEY KOREN, ARANG RHIE, MIKKO RAUTIAINEN, ANDREY V. BZIKADZE, ALLA MIKHEENKO, MITCHELL R. VOLLGER, NICOLAS ALTEMOSE, LEV URALSKY, [...] AND ADAM M. PHILLIPPY, +90 authors [Authors Info & Affiliations](#)

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T2T consortium made heavy use of PacBio and ONT long reads

# Base calling: the conversion of signal to a nucleotide sequence



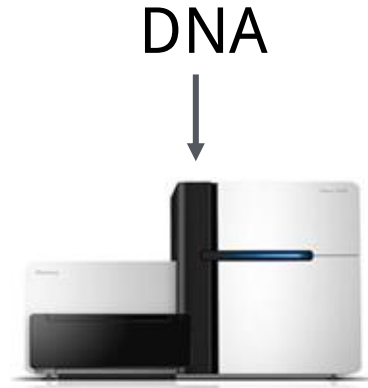
Raw signal  
(e.g., 454 Life Sciences)

Errors happen.  
Hopefully infrequently

Base calling algorithms

ACCTTCGAACGGCGGGGGGTTACAA

(Mostly) all technologies yield DNA sequences in FASTQ format



@seq1  
ACCTTCGAACGGCGGGGGGTTACAA  
+  
!''\*((( (\*\*\*) )%%%++) .1\*\*\*  
@seq2  
TGGAACCGAACGGCCCCGGTTACAT  
+  
!''\*!!!! (\*\*\*) )+++++ ) .1\*\*\*  
And so on...

# The FASTQ format. Welcome to a minor hell.

A “standard” format for storing and defining sequences from next-generation sequencing technologies.

```
Sequence ID @SEQ_ID
Sequence GATTTGGGGTTCAAAGCAGTATCGATCAAATAGTAAATCCATTTGTTCAACTCACAGTTT
<separator> +
Quality scores !''*(((((***+))%%%++) (%%%) .1***-+*'')) **55CCF>>>>>CCCCCCC65
```

[http://en.wikipedia.org/wiki/FASTQ\\_format](http://en.wikipedia.org/wiki/FASTQ_format)

# The FASTQ format's sequence identifier (first line of each record)

## Old format

```
@HWUSI-EAS100R:6:73:941:1973#0/1
```

|                      |                                                                              |
|----------------------|------------------------------------------------------------------------------|
| <b>HWUSI-EAS100R</b> | the unique instrument name                                                   |
| <b>6</b>             | flowcell lane                                                                |
| <b>73</b>            | tile number within the flowcell lane                                         |
| <b>941</b>           | 'x'-coordinate of the cluster within the tile                                |
| <b>1973</b>          | 'y'-coordinate of the cluster within the tile                                |
| <b>#0</b>            | index number for a multiplexed sample (0 for no indexing)                    |
| <b>/1</b>            | the member of a pair, /1 or /2 ( <i>paired-end or mate-pair reads only</i> ) |

## New format

```
@EAS139:136:FC706VJ:2:2104:15343:197393 1:Y:18:ATCACG
```

|                |                                                                            |
|----------------|----------------------------------------------------------------------------|
| <b>EAS139</b>  | the unique instrument name                                                 |
| <b>136</b>     | the run id                                                                 |
| <b>FC706VJ</b> | the flowcell id                                                            |
| <b>2</b>       | flowcell lane                                                              |
| <b>2104</b>    | tile number within the flowcell lane                                       |
| <b>15343</b>   | 'x'-coordinate of the cluster within the tile                              |
| <b>197393</b>  | 'y'-coordinate of the cluster within the tile                              |
| <b>1</b>       | the member of a pair, 1 or 2 ( <i>paired-end or mate-pair reads only</i> ) |
| <b>Y</b>       | Y if the read is filtered, N otherwise                                     |
| <b>18</b>      | 0 when none of the control bits are on, otherwise it is an even number     |
| <b>ATCACG</b>  | index sequence                                                             |

# FASTQ quality scores: estimate of confidence in each base (sequencing technologies make errors!)

|                |                                                                                                                       |
|----------------|-----------------------------------------------------------------------------------------------------------------------|
| Sequence ID    | @SEQ_ID                                                                                                               |
| Sequence       | GATTTGGGGTTCAAAGCAGTATCGATCAAATAGTAAATCCATTTGTTCAACTCACAGTTT                                                          |
| <separator>    | +                                                                                                                     |
| Quality scores | ! ' ' * ( ( ( * * * + ) ) % % % + + ) ( % % % % ) . 1 * * * - + * ' ' ) ) * * 5 5 C C F > > > > > C C C C C C C C 6 5 |

# FASTQ quality scores: estimate of confidence in each base (sequencing technologies make errors!)

|                |                                                              |
|----------------|--------------------------------------------------------------|
| Sequence ID    | @SEQ_ID                                                      |
| Sequence       | GATTTGGGGTTCAAAGCAGTATCGATCAAATAGTAAATCCATTTGTTCAACTCACAGTTT |
| <separator>    | +                                                            |
| Quality scores | !''*(((((***+))%%%+)) (%%%).1***-+*''))**55CCF>>>>>CCCCCCC65 |



Qualities are based on the Phred scale and are *encoded*

$$Q = -10 \cdot \log_{10}(P_{\text{err}})$$

# Phred quality score calculation

$$Q = -10 \cdot \log_{10}(P_{\text{err}})$$

| Error probability<br>( $P_{\text{err}}$ ) | $\log_{10}(P_{\text{err}})$ | Phred quality<br>score |
|-------------------------------------------|-----------------------------|------------------------|
| 1                                         | 0                           | 0                      |
| 0.1                                       | -1                          | 10                     |
| 0.01                                      | -2                          | 20                     |
| 0.001                                     | -3                          | 30                     |
| 0.0001                                    | -4                          | 40                     |

# FASTQ quality scores: estimate of confidence in each base (sequencing technologies make errors!)

|                |                                                                                                                      |
|----------------|----------------------------------------------------------------------------------------------------------------------|
| Sequence ID    | @SEQ_ID                                                                                                              |
| Sequence       | GATTTGGGGTTCAAAGCAGTATCGATCAAATAGTAAATCCATTGTTCAACTCACAGTT                                                           |
| <separator>    | +                                                                                                                    |
| Quality scores | ! ' ' * ( ( ( * * * + ) ) % % % + + ) ( % % % % ) . 1 * * * -<br>+ * ' ' ) * * 5 5 C C F > > > > > C C C C C C C 6 5 |

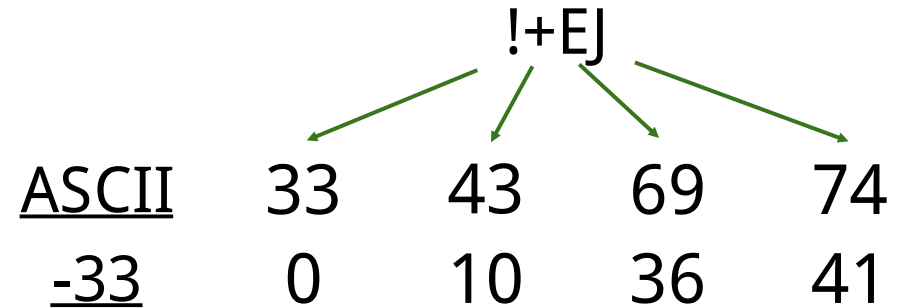
# Quality score encoding based on ASCII table

| Dec | Hex | Char             | Dec | Hex | Char  | Dec | Hex | Char | Dec | Hex | Char |
|-----|-----|------------------|-----|-----|-------|-----|-----|------|-----|-----|------|
| 0   | 00  | Null             | 32  | 20  | Space | 64  | 40  | @    | 96  | 60  | `    |
| 1   | 01  | Start of heading | 33  | 21  | !     | 65  | 41  | A    | 97  | 61  | a    |
| 2   | 02  | Start of text    | 34  | 22  | "     | 66  | 42  | B    | 98  | 62  | b    |
| 3   | 03  | End of text      | 35  | 23  | #     | 67  | 43  | C    | 99  | 63  | c    |
| 4   | 04  | End of transmit  | 36  | 24  | \$    | 68  | 44  | D    | 100 | 64  | d    |
| 5   | 05  | Enquiry          | 37  | 25  | %     | 69  | 45  | E    | 101 | 65  | e    |
| 6   | 06  | Acknowledge      | 38  | 26  | &     | 70  | 46  | F    | 102 | 66  | f    |
| 7   | 07  | Audible bell     | 39  | 27  | '     | 71  | 47  | G    | 103 | 67  | g    |
| 8   | 08  | Backspace        | 40  | 28  | (     | 72  | 48  | H    | 104 | 68  | h    |
| 9   | 09  | Horizontal tab   | 41  | 29  | )     | 73  | 49  | I    | 105 | 69  | i    |
| 10  | 0A  | Line feed        | 42  | 2A  | *     | 74  | 4A  | J    | 106 | 6A  | j    |
| 11  | 0B  | Vertical tab     | 43  | 2B  | +     | 75  | 4B  | K    | 107 | 6B  | k    |
| 12  | 0C  | Form feed        | 44  | 2C  | ,     | 76  | 4C  | L    | 108 | 6C  | l    |
| 13  | 0D  | Carriage return  | 45  | 2D  | -     | 77  | 4D  | M    | 109 | 6D  | m    |
| 14  | 0E  | Shift out        | 46  | 2E  | .     | 78  | 4E  | N    | 110 | 6E  | n    |
| 15  | 0F  | Shift in         | 47  | 2F  | /     | 79  | 4F  | O    | 111 | 6F  | o    |
| 16  | 10  | Data link escape | 48  | 30  | 0     | 80  | 50  | P    | 112 | 70  | p    |
| 17  | 11  | Device control 1 | 49  | 31  | 1     | 81  | 51  | Q    | 113 | 71  | q    |
| 18  | 12  | Device control 2 | 50  | 32  | 2     | 82  | 52  | R    | 114 | 72  | r    |
| 19  | 13  | Device control 3 | 51  | 33  | 3     | 83  | 53  | S    | 115 | 73  | s    |
| 20  | 14  | Device control 4 | 52  | 34  | 4     | 84  | 54  | T    | 116 | 74  | t    |
| 21  | 15  | Neg. acknowledge | 53  | 35  | 5     | 85  | 55  | U    | 117 | 75  | u    |
| 22  | 16  | Synchronous idle | 54  | 36  | 6     | 86  | 56  | V    | 118 | 76  | v    |
| 23  | 17  | End trans. block | 55  | 37  | 7     | 87  | 57  | W    | 119 | 77  | w    |
| 24  | 18  | Cancel           | 56  | 38  | 8     | 88  | 58  | X    | 120 | 78  | x    |
| 25  | 19  | End of medium    | 57  | 39  | 9     | 89  | 59  | Y    | 121 | 79  | y    |
| 26  | 1A  | Substitution     | 58  | 3A  | :     | 90  | 5A  | Z    | 122 | 7A  | z    |
| 27  | 1B  | Escape           | 59  | 3B  | ;     | 91  | 5B  | [    | 123 | 7B  | {    |
| 28  | 1C  | File separator   | 60  | 3C  | <     | 92  | 5C  | \    | 124 | 7C  |      |
| 29  | 1D  | Group separator  | 61  | 3D  | =     | 93  | 5D  | ]    | 125 | 7D  | }    |
| 30  | 1E  | Record separator | 62  | 3E  | >     | 94  | 5E  | ^    | 126 | 7E  | ~    |
| 31  | 1F  | Unit separator   | 63  | 3F  | ?     | 95  | 5F  | _    | 127 | 7F  |     |

Formula for getting PHRED quality from encoded quality:

$$Q = \text{ascii}(\text{char}) - 33$$

Example:



# Historically, FASTQ has had different encoding schemes for encoding PHRED quality scores. Ouch.



Current encoding:  
! = quality 0  
J = quality 41

# FASTQE

<https://github.com/fastqe/fastqe>

```
$ fastqe example.fastq --min --max --bin
```

```
example.fastq      max (binned)
```



```
example.fastq      mean (binned)
```



```
example.fastq      min (binned)
```



# Quality score binning

**Table 1: Q-Score Bins for an Optimized 8-Level Mapping**

| <b>Quality Score Bins</b> | <b>Example of Empirically Mapped Quality Scores*</b> |
|---------------------------|------------------------------------------------------|
| N (no call)               | N (no call)                                          |
| 2–9                       | 6                                                    |
| 10–19                     | 15                                                   |
| 20–24                     | 22                                                   |
| 25–29                     | 27                                                   |
| 30–34                     | 33                                                   |
| 35–39                     | 37                                                   |
| ≥ 40                      | 40                                                   |

# FASTQ vs FASTA

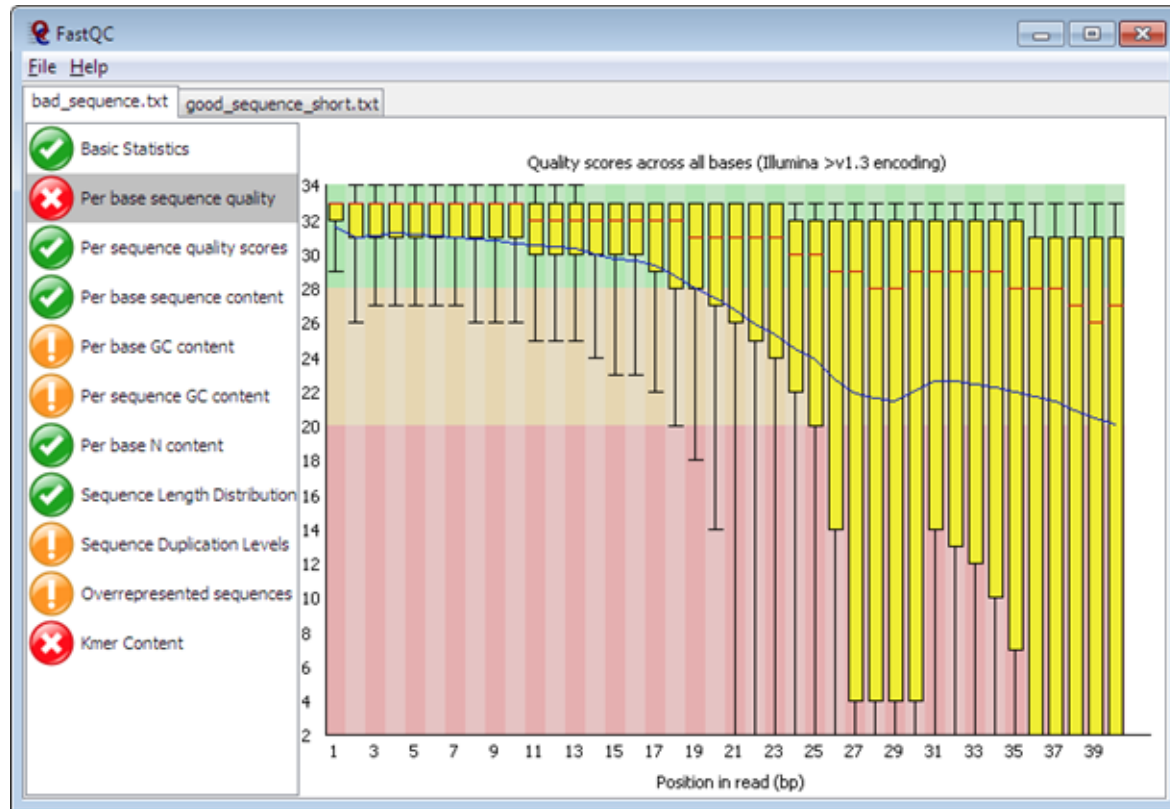
>seq1

```
ACGACGAGACCTTCATCAAAAACATCATCATCCAGGACTGTATGTGGAGCGGCTTCTCGG
CCGCCGCCAAGCTCGTCTCAGAGAAGCTGGCCTCCTACCAGGCTGCGCGCAAAGACAGCG
GCAGCCCGAACCCCGCCCGCGGCCACAGCGTCTGCTCCACCTCCAGCTTGTACCTGCAGG
ATCTGAGCGCCGCGCCTCAGAGTGCATCGACCCCTCGGTGGTCTTCCCCTACCCTCTCA
ACGACAGCAGCTCGCCCAAGTCCTGCGCCTCGCAAGACTCCAGCGCCTTCTCTCCGTCCT
CGGATTCTCTGCTCTCCTCGACGGAGTCCTCCCCGCAGGGCAGCCCCGAGC
```

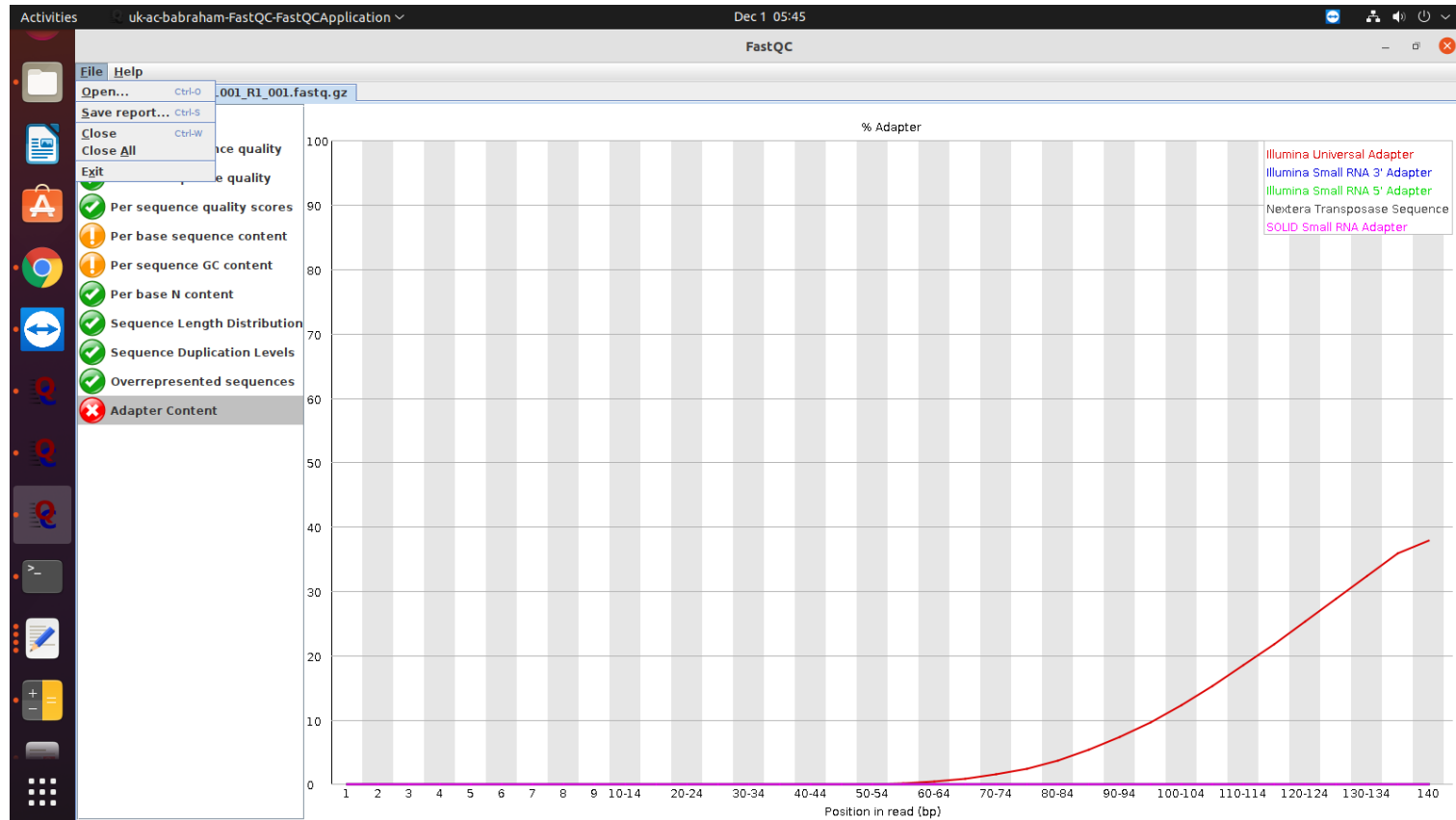
>seq2

```
TCCATGAGGAGACACCGCCCACCACCAGCAGCGACTCTGGTAAGCGAAGCCCGCCCAGGC
CTGTCAAAAGTGGGCGGCTGGATACCTTTCCCATTTTCATTGGCAGCTTATTTAACGGGC
CACTCTTATTAGGAAGGAGAGATAGCAGATCTGGAGAGATTTGGGAGCTCATCACCTCTG
AAACCTTGCGCTTTAGCGTTTCCTCCCATCCCTTCCCCTTAGACTGCCCATGTTTGCAGC
CCCCCTCCCCGTTTGTCTCCACCCCTCAGGAATTTCAATTTAGGTTTTTAAACCTTCTGG
CTTATCTTACAACCTCAATCCACTTCTTCTTACCTCCCGTTAACATTTTAATTGCCCTGGG
GCGGGGTGGCAGGGAGTGTATGAATGAGGATAAGAGAGGATTGATCTCTGAGAGTGAATG
AATTGCTTCCCTCTTAACCTCCGAGAAGTGGTGGGATTTAATGAACTATCTACAAAAATG
```

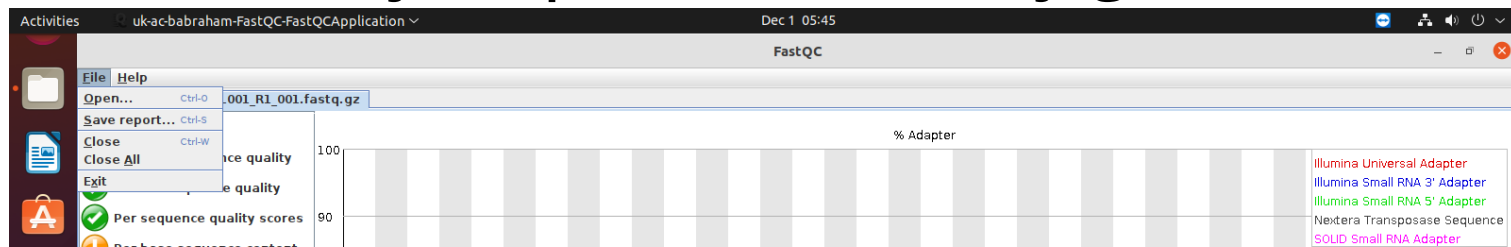
# FASTQC: Is my sequence data any good?



# FASTQC: Is my sequence data any good?



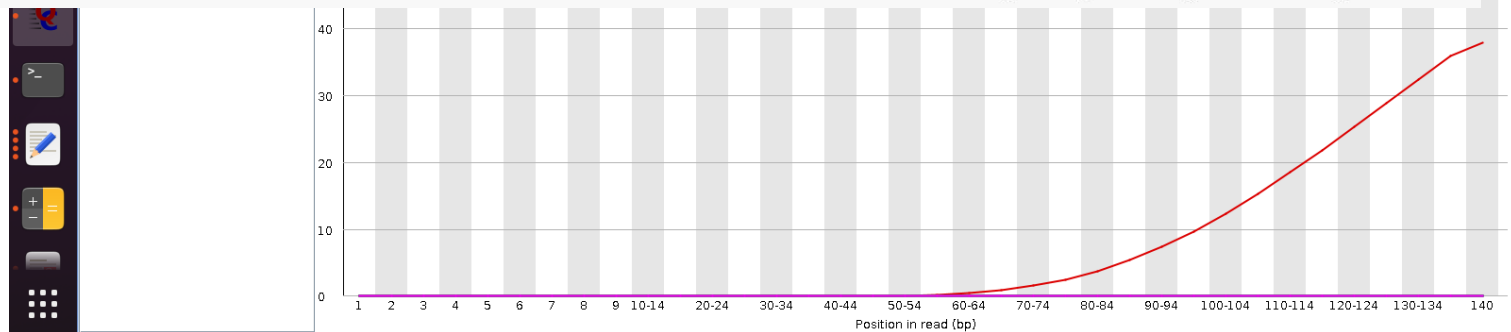
# FASTQC: Is my sequence data any good?



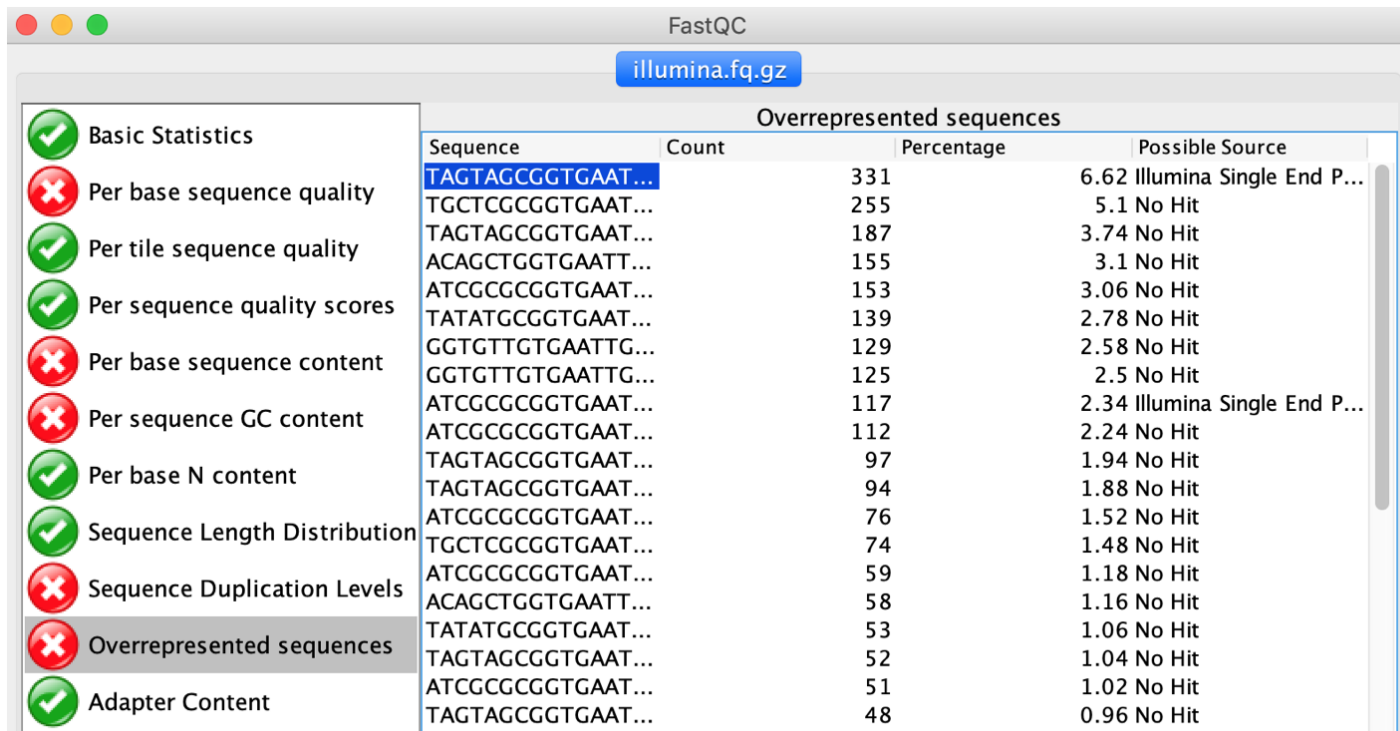
(good): [adapter]----- insert -----[adapter]  
read 1 ----->

(bad): [adapter]---- insert ----[adapter]  
read 1 ----->

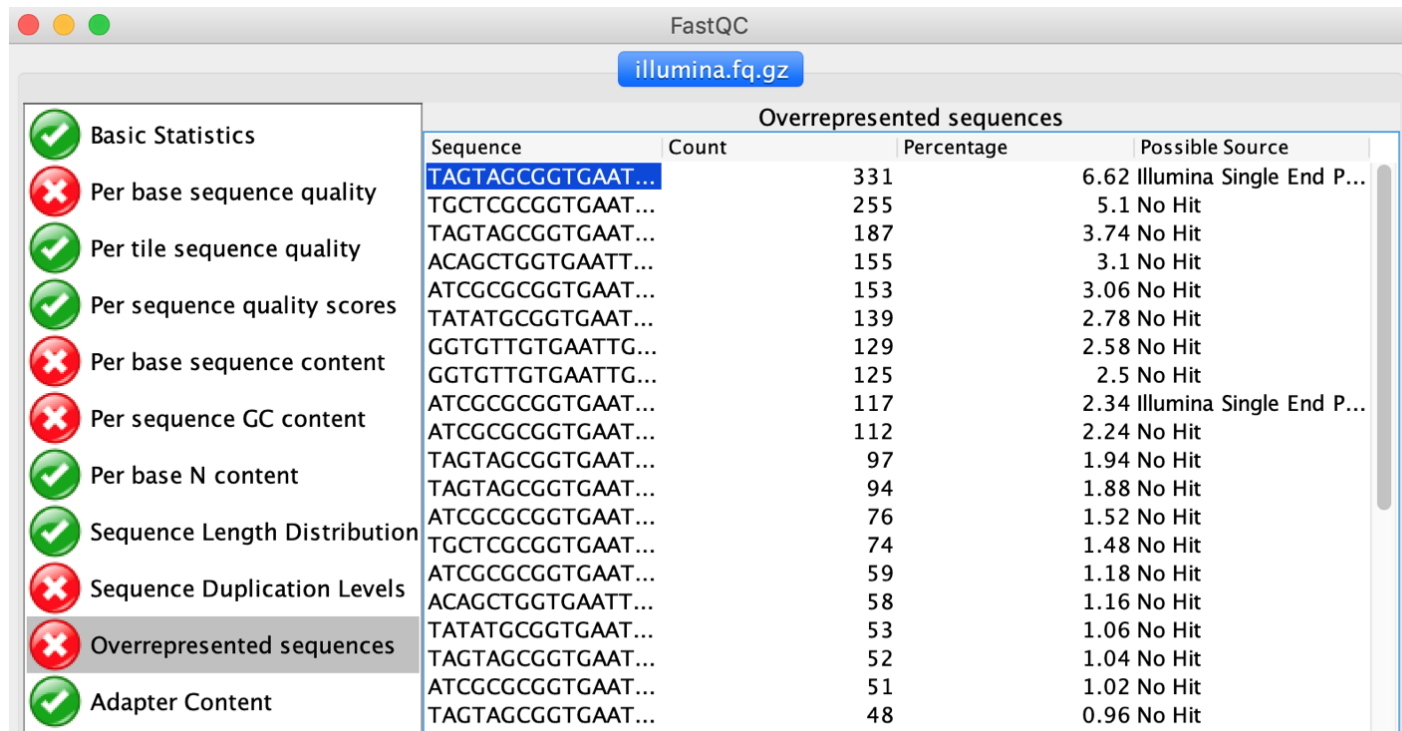
^^^^^ these bases are adapter, not your sample



# FASTQC: Is my sequence data any good?

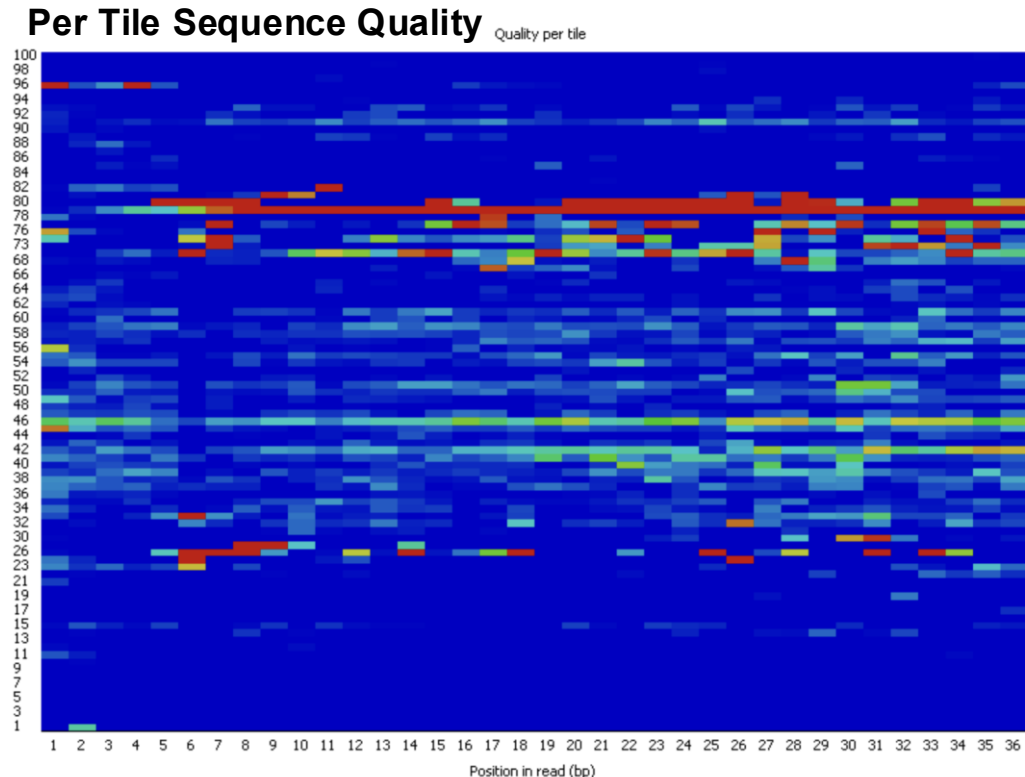


# FASTQC: Is my sequence data any good?



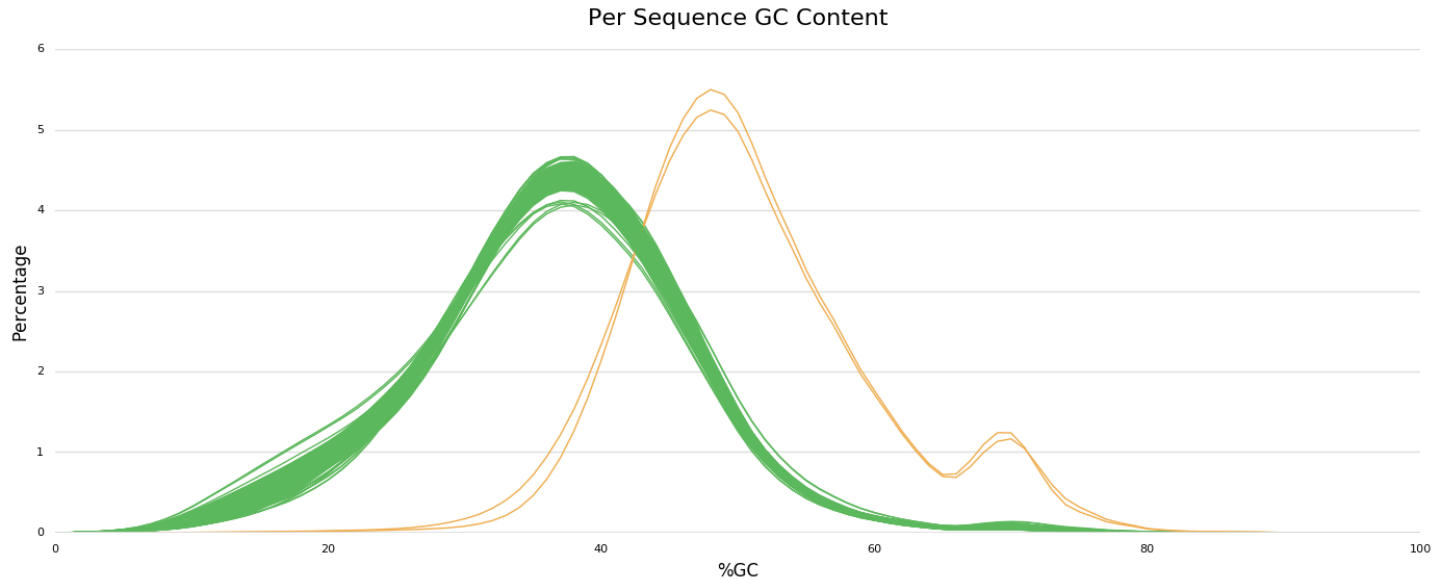
RNA-seq  
Amplikon sequencing  
Gene panels, deeply sequenced

# FASTQC: Is my sequence data any good?

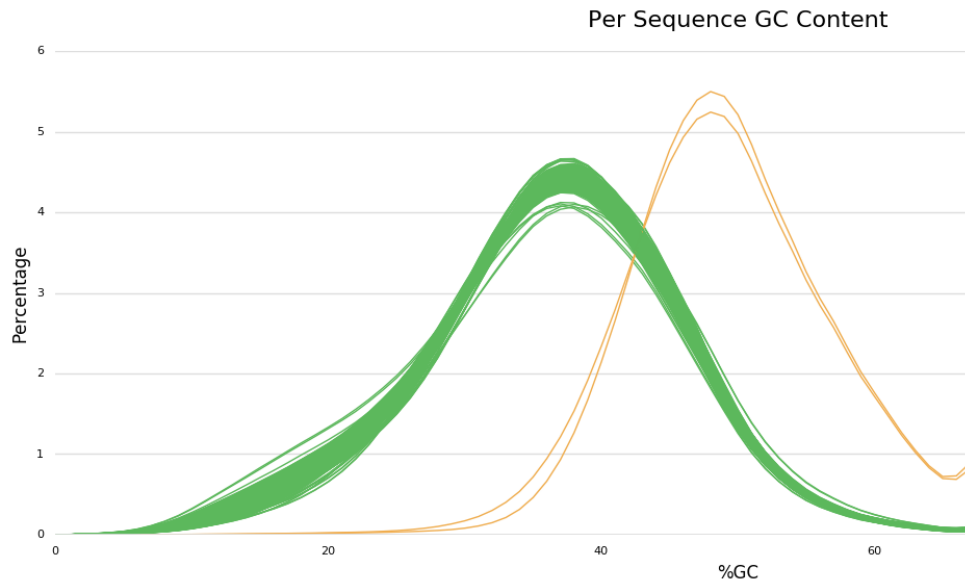


<https://www.bioinformatics.babraham.ac.uk/projects/fastqc/Help/>

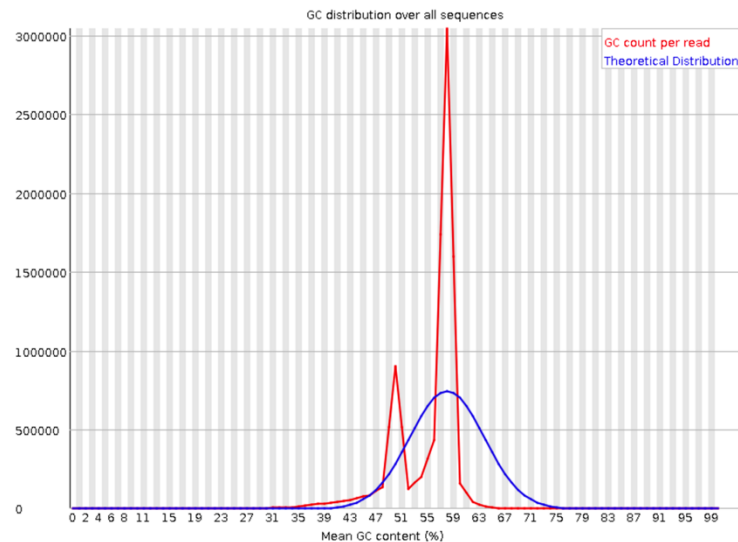
# FASTQC: Is my sequence data any good?



# FASTQC: Is my sequence data any good?



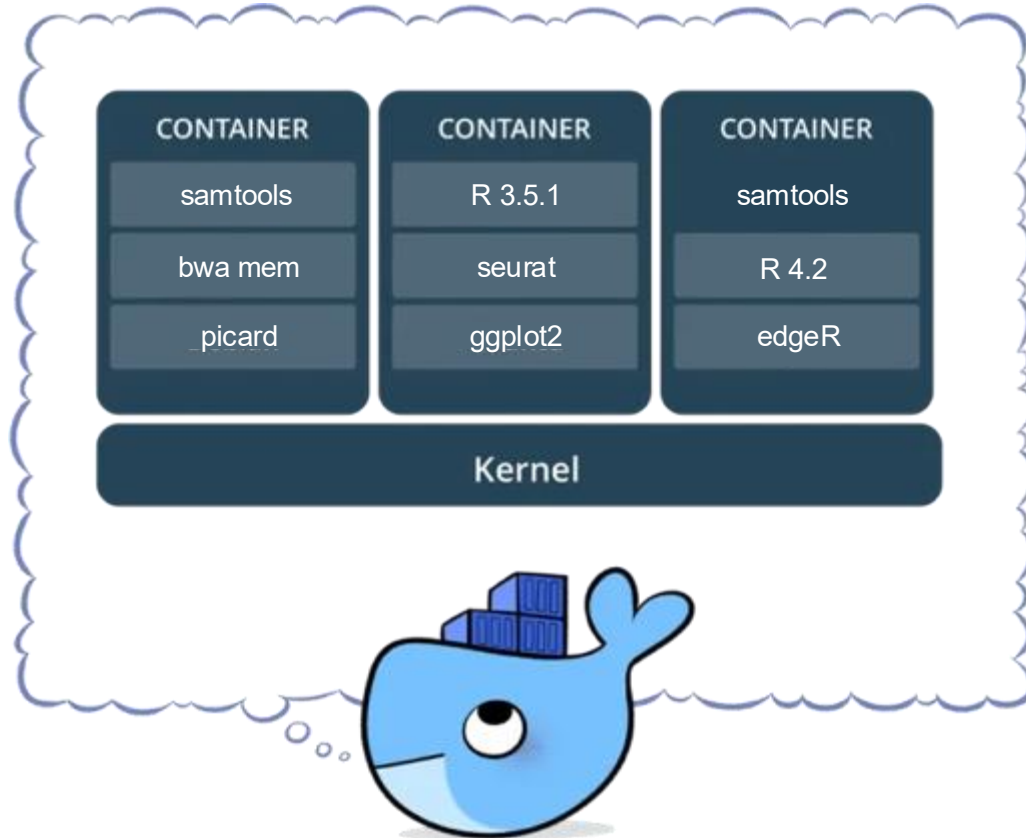
✖ Per sequence GC content



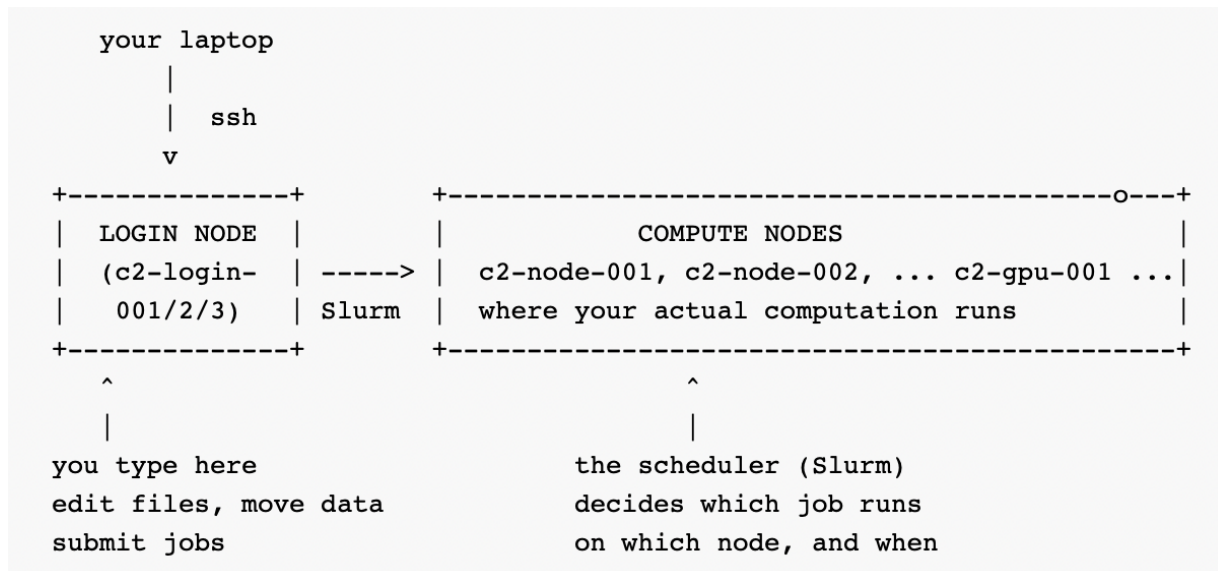
# Computing Environments

- Laptop
  - You administer
  - You control completely
- Shared compute cluster
  - A sysadmin or group administers it
  - You control very little
- Docker (containers)
  - Sysadmins handle the hardware
  - You control the software almost completely

# Docker containers

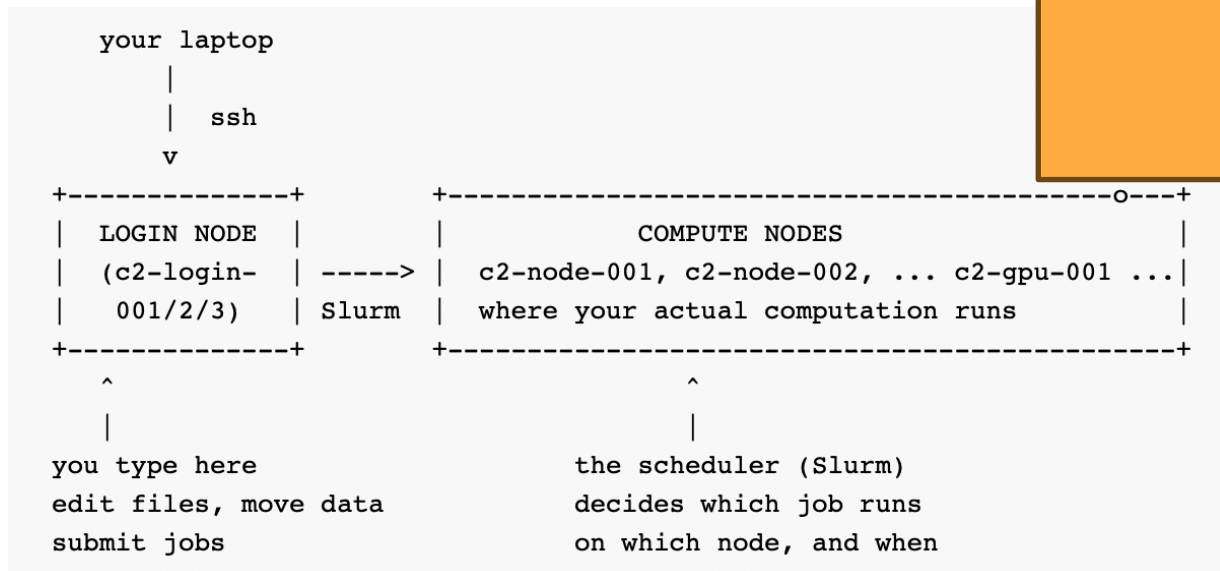


# Docker containers

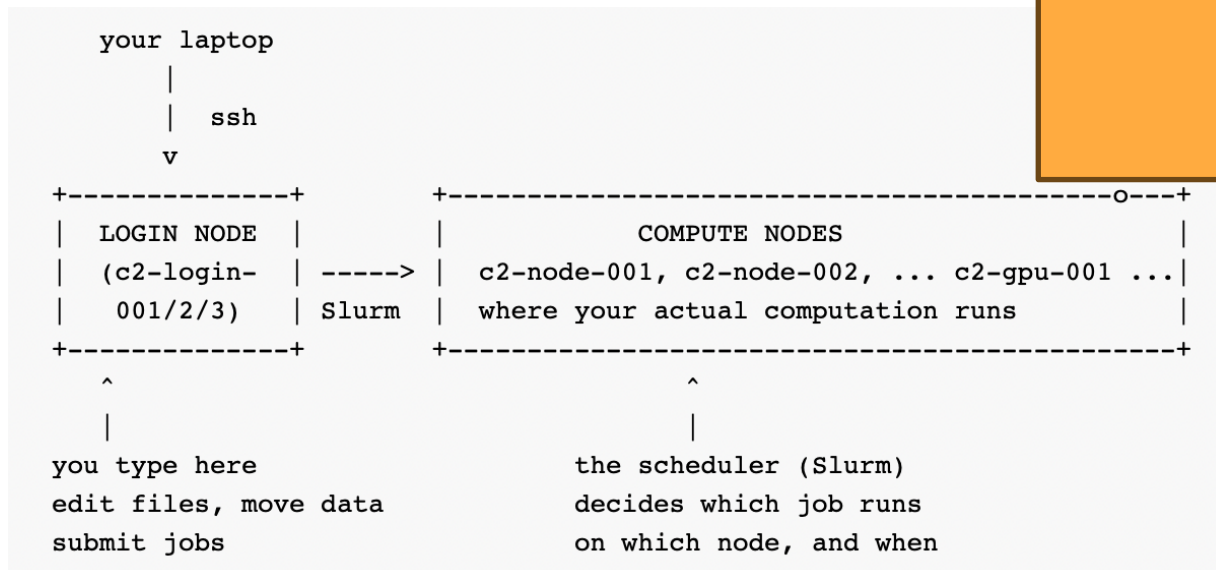
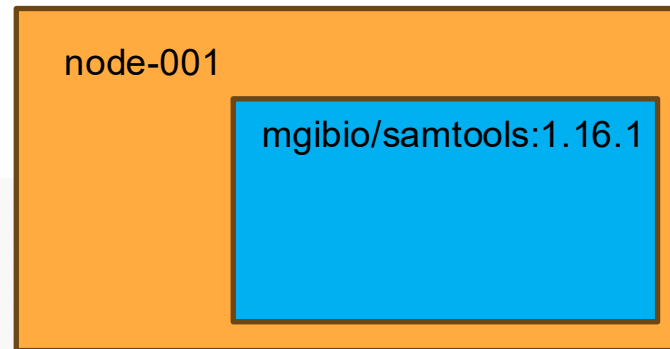


# Docker containers

node-001



# Docker containers



# Docker benefits

- Many different software environments, one cluster
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  - Hell is other people's data

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  - Hell is your own data, 6 months later